# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-SEXAGESIMAL-EN · v1.5 · 2026-10-20 |
| Package | star-sexagesimal |
| Version | 0.1.0 |
| Capability | star.units.sexagesimal |
| Maturity | RELEASED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_sexagesimal |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())"
star_sexagesimal-0.1.0-py3-none-any.whl
```

| FILE | SHA-256 |
| --- | --- |
| star_sexagesimal-0.1.0-py3-none-any.whl | 6e28b6c3a6dc04b5c21050e124b9e9216c67f925fa491678d8288a20aea6967d |
| star_sexagesimal.py | 07e835372095a413980d157be650a4c907b9af3fd37a99ce6c26fac087e4b6ed |
| star_sexagesimal-0.1.0-source.zip | 7d317ad86e47314fcbf1930135e26cf3fa5e7c082790be1252e4b8526c4b95dc |

# 5. What it does, and its limits

Degrees to and from degrees/minutes/seconds and hours/minutes/seconds with one rounding on the whole angle; rounded fields identical to ERFA on 1200 of 1200 cases, joined angles within 1.2e-13 deg of ERFA and astropy, and out-of-range input refused.

No formatting to text and no parsing of text: fields in, fields out. No wrapping into [0, 360) or [0, 24 h). At most six decimals of a second. A binary floating-point angle that is not exactly on a tie is rounded as the float it is, not as the decimal it was typed as.

- `deg_to_dms(deg, decimals)` and `deg_to_hms(deg, decimals)` return (sign, whole, minutes, seconds, fraction) as a '+'/'-' sign and four integers, `fraction` in units of 10**-decimals second; rounding to the nearest unit, ties away from zero, once on the whole angle. Reproduce 23° 26' 21.448" and 17h 48m 59.74s (Meeus).
- `dms_to_deg` and `hms_to_deg` join the fields; 15° = 1 h. No wrapping: 360° is 360° and 24 h.
- Measured on 600 angles (100 within 1e-9 deg of a whole arcminute) and 600 sets of fields: rounded fields identical to ERFA in 1200 of 1200, never a 60; joined angle within 1.2e-13 deg of ERFA and astropy; unrounded seconds within 5e-7 s of astropy (the rounding step at six decimals).
- Every function returns its result or raises `ValueError`: non-numeric, boolean, NaN or infinite input, an angle outside [-360, 360], decimals not an integer in 0...6, a sign other than '+' or '-', degrees not an integer in 0...360 (hours 0...24), minutes not an integer in 0...59, seconds outside [0, 60), a total above 360° (24 h).

## 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

## 7. Citation and bill of materials

```
@software{cavazzini_star_sexagesimal_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-sexagesimal}: Degrees ⇄ degrees/minutes/seconds and
hours/minutes/seconds},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_sexagesimal/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

## 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts